

Color Recognition System Based on Arduino Nano with Voice Output as an Assistive Tool for the Visually Impaired

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Abstract— Color identification is a daily challenge for individuals with visual impairments, especially when selecting clothing or recognizing colored object. This research proposes a simple and low-cost color recognition system using an Arduino Nano, designed to provide voice-based feedback through a speaker. The system detects color using a TCS34725 sensor and classifies them using a lightweight algorithm, giving real-time audio output to the user. Testing shows the system can accurately recognize basic colors. With its compact and accessible design, the device serves as a practical tool to assist visually impaired users in identifying colors independently.

Keywords—Color Recognition, Visual Impairment, Arduino Nano, Assistive Technology, Voice Output

I. INTRODUCTION

Color plays a pivotal role in daily life, aiding in object identification, clothing selection, and interpreting environmental cues. For individuals with visual impairments, distinguishing colors poses significant challenges, impacting their autonomy and quality of life [1]. Traditional assistive tools, such as white canes or braille labels, offer limited support in conveying color information.

Advancements in technology have paved the way for assistive that bridge this gap. The integration of color sensors, microcontrollers, and audio output modules has become more accessible and affordable, enabling the development of portable color recognition tools tailored for visually impaired users [2]. These system aim to detect object colors and convert the information into alternative formats, such as auditory feedback, to aid perception [3].

In this research, we present a compact, voice-based color recognition device utilizing the Arduino Nano microcontroller, the TCS34725 color sensor, and the DFPlayer Mini Audio module. The system is designed to detect the color of an object upon button activation and provide immediate voice feedback through a speaker. To classify colors accurately, a lightweight k-Nearest Neighbors (k-NN) algorithm is implemented, allowing the device to recognize basic and commonly encountered colors in everyday life [4].

This The primary contributions of this study include the integration of cost-effective hardware components, the application of a simple yet effective classification method, and a focus on user-friendly design for visually impaired individuals. Powered by the 18650 lithium-ion batteries, the

device ensures portability and extended operation without reliance on external power sources.

By offering an affordable and easy-to-use solution, this research aims to enhance the independence of visually impaired individuals in identifying colors across various real-life scenarios.

II. METHODOLOGY

A. Research Design

This research adopts an experimental and engineering design approach, aimed at developing a colour recognition assistive device for visually impaired users. The device is capable of detecting the colour of objects through a button-activated process and provides voice feedback based on the detected colour class. The system process RGB data obtained from a TCS34725 sensor, converts it to HSV colour space, normalizes the values, and performs classification using a simple k-Nearest Neighbours (k-NN) algorithm implemented directly on the Arduino Nano microcontroller.

B. Hardware Components

The main components used in this system include:

- **Arduino Nano:** Functions as the main controller due to its compact size and compatibility with basic classification logic [5].
- **TCS34725 RGB Colour Sensor:** A digital sensor with integrated IR filter and white LED, capable of accurately capturing ambient color in RGB format [6].
- **DFPlayer Mini MP3 Module:** Used for audio output, capable of playing pre-recorded colour names stored on a micro SD card [7].
- **Push Button:** Serves as a trigger for the colour detection process.
- **Speaker (8Ω):** Outputs the detected colour name via audio feedback.
- **Battery Supply:** Two 18650 batteries connected via a protection module provide mobile power to the system.

This configuration allows the device to function independently and portably, suitable for everyday use in identifying object colours.

C. System Operation

The operation flow of the system is as follows:

1. The user presses a push button to initiate the detection process.
2. The TCS34725 sensor reads the RGB values of the object in front of it.
3. The system converts RGB values to HSV using standard conversion formulas [8].
4. The HSV values are normalized between 0 and 1 for consistent comparison.
5. A simple k-NN (k=1) algorithm compares the normalized HSV input with a pre-defined dataset of labelled color values.
6. The nearest match is selected based on Euclidian distance, and the corresponding audio file (e.g., "red.mp3") is played using DFPlayer Mini.
7. The system returns to idle state, waiting for the next button press.

This design provides an accessible and intuitive user experience for colour recognition.

D. Data Processing and Algorithm

a. RGB to HSV conversion

RGB values obtained from the sensor are first converted to HSV using the following transformation [8]:

- Normalize R, G, B to the range [0, 1]
- Compute $Max = \max(R, G, B)$
- Compute $Min = \min(R, G, B)$
- $V = Max$
- $S = \frac{(Max - Min)}{Max}$ (if $Max \neq 0$)
- Hue (H) is calculated based on which color is Max:
 - If $Max = R: H = 60 \times \frac{(G - B)}{Max - Min}$
 - If $Max = G: H = 60 \times (2 + \frac{B - R}{Max - Min})$
 - If $Max = B: H = 60 \times (4 + \frac{(R - G)}{Max - Min})$

b. Normalization

To reduce variation in input readings, HSV values are normalized to a 0-1 range: $x' = \frac{x - x_{min}}{x_{max} - x_{min}}$

Where x is each HSV component, and x_{min}, x_{max} are min/max values defined based on expected sensor readings.

c. k-NN Classification

A basic 1-nearest neighbor classifier compares the input HSV value to a dataset using Euclidian distance:

$$d = \sqrt{(H_1 - H_2)^2 + (S_1 - S_2)^2 + (V_1 - V_2)^2}$$

The label of the nearest known data point is selected as the recognized color [9].

The pre-labeled dataset is stored in code as an array of HSV values and corresponding audio file indices (e.g., {H,S,V,"red.mp3"}).

E. Block Diagram

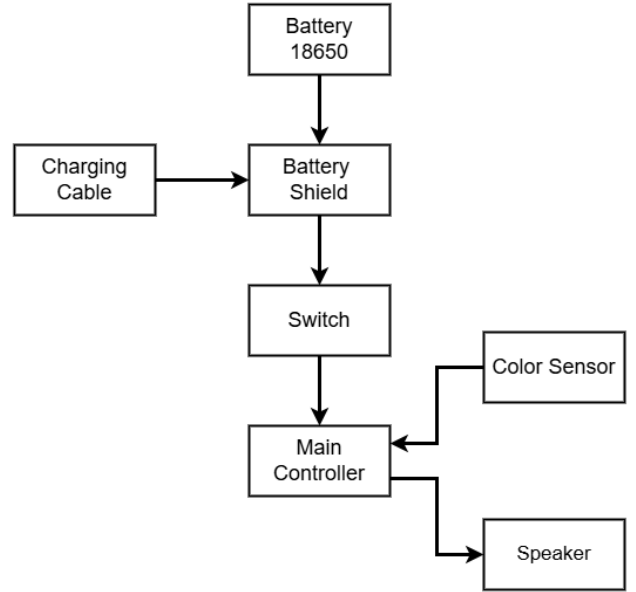


Fig. 1. Color Recognition Block Diagram.

The block diagram of the color recognition tool illustrates the interaction between key hardware components. The core of the system is the Arduino Nano microcontroller, which controls the flow of data and logic. The TCS34725 color sensor captures RGB color data from the object when the push button is pressed. The captured RGB values are then processed inside the microcontroller using a simple k-NN classification algorithm to determine the most likely color label.

Once the color has been classified. The system converts the label into voice feedback using the DFPlayer Mini Module, which plays a corresponding pre-recorded audio file through a speaker. The entire system is powered by two 18650 lithium batteries connected via a power module. This configuration allows the device to operate portably and without needing external input or display, making it suitable for visually impaired users.

F. Flowchart

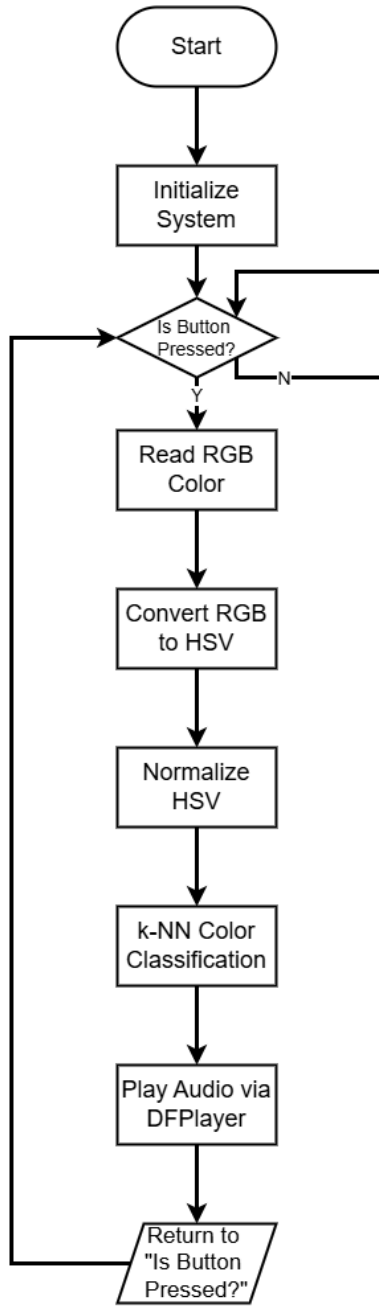


Fig. 2. Color Recognition Flowchart.

The flowchart present the main operational steps of the colour recognition tool. The process begins by waiting for the user to press the button, which serves as a trigger for colour recognition. Once the button is pressed, the system reads RGB data from the sensor, converts the RGB values to HSV, and then normalizes them. Using the normalized values, the system applies a k-NN algorithm to classify the colour.

The identified color label is then used to play corresponding audio file via the DFPlayer Mini. After the audio is played, the system returns to the initial state, waiting for the next button press. If the button is not pressed, the system, the system remains in an idle loop. This structure ensures that the tool only activates when needed and minimizes power consumption during standby.

III. RESULT AND DISCUSSION

The developed colour recognition device was tested under different lighting conditions and object distance to evaluate its accuracy in colour classification. Additional analysis was conducted to examine sensor reading consistency at 0cm and 3cm, especially in challenging colour such as black, grey, and white. Observations on the device's physical form and practical usability were also made to support a more holistic of its performance.

A. Device Appearance

The developed colour recognition device is enclosed in a compact 3D-printed casing with approximate dimensions of 10cm x 6cm x 5.5cm. The casing is made of PLA filament, selected for its lightweight and ease of fabrication, making the device both durable and portable. The size is designed to strike a balance between portability and usability.

On the exterior of the device, several functional components are visibly accessible. At the front dedicated opening aligns with the TCS34725 colour sensor, allowing direct contact-based colour detection from the object surface. A push button is placed on the top of the device to initiate the colour detection process. A power button is installed to turn the device on. In addition, the casing includes small openings for a speaker, used to output the detected colour via audio, and USB ports (micro USB and USB-C), which are provided for charging the internal battery.

All electronic components such as the microcontroller, sensor modules, audio player, and battery system are housed securely inside the enclosure, hidden from view.



Fig. 3. Laser Vision Device.

TABLE I. RGB SENSOR BRIGHT EXPERIMENT RESULTS

Lighting	Distance (cm)	Matches / Total	Average Error
Bright	0	13 / 16	81.25%
Bright	1	5 / 16	31.25%
Bright	2	2 / 16	12.5%

B. Training Data Consideration

To ensure optimal color classification using the k-Nearest Neighbours (k-NN) algorithm, the selection of training data plays critical role. In this research, training data were collected by placing the colour sensor at a distance of 0cm from the surface of the colored object. This decision was based on practical considerations—specifically, the design of the device casing places the sensor in direct contact with the target object during operation. Data collected at 0cm were considered the most consistent and accurate for matching real-world usage conditions.

Prior to settling on 0cm, training data were also experimentally collected at a distance of 3cm, and a sensor tube was temporarily added to reduce ambient light interference. However, the results from this configuration exhibited less consistency due to greater variation in lighting and sensor angle. Additionally, HSV readings for the same colour at 3cm differed significantly from those at 0cm, indicating that proximity plays a major role in colour accuracy.

The following table outlines an example of HSV sensor readings for selected object colours at 0cm and 3cm, showing the variation that supports the decision to train the model using 0cm data.

TABLE II. COLOR DETECTION ACCURACY

Object Color	HSV at 0cm	HSV at 3cm
Sulfur Black	46, 0.38, 0.40	73, 0.32, 0.38
Chili Red	1, 0.81, 0.83	8, 0.68, 0.62
Pure White	43, 0.44, 0.42	40, 0.45, 0.42
Fuji Green	132, 0.51, 0.50	105, 0.36, 0.43
Canary Yellow	39, 0.69, 0.49	37, 0.66, 0.48
Light Turquoise	138, 0.37, 0.43	123, 0.28, 0.41
Bright Royal Blue	202, 0.27, 0.42	116, 0.14, 0.38
Magenta	1, 0.68, 0.69	19, 0.50, 0.49

The results reinforce that sensor-to-object proximity is critical for consistent colour detection. Training at 0cm provides better alignment with real usage, especially considering that in the final device, the sensor is positioned inside a fixed casing with close-range contact. Future development may consider dynamic distance calibration or training at multiple distances to enhance flexibility.

At 1cm under dim lighting, accuracy was higher than in the bright environment, with 11 out of 16 matches (68.75%). This unexpected improvement might be due to reduced glare or surface reflection in the darker environment, which can affect color reading at slight distances.

C. Color Detection Accuracy Analysis

This section presents the analysis of the device’s accuracy in detecting object colors under varying conditions. The test involved comparing the detected color from the device with the actual object color (ground truth) at different distances and lighting levels. Each detection was classified as a match if color detected was visually similar to the actual color as judged by human perception, even if the color names differed slightly.

Five test scenarios were conducted:

- Bright lighting at 0cm
- Bright lighting at 1cm
- Bright lighting at 2cm
- Dim lighting at 0cm
- Dim lighting at 1cm

The results are summarized in the following table:

TABLE III. COLOR DETECTION ACCURACY

Lighting	Distance (cm)	Matches / Total	Accuracy (%)
Bright	0	13 / 16	81.25%
Bright	1	5 / 16	31.25%
Bright	2	2 / 16	12.5%
Dim	0	13 / 16	81.25%
Dim	1	11 / 16	68.75%

The results show that the most optimal performance occurs at 0cm, both under bright and dim lighting, with identical accuracy of 81.25%. This consistency is likely due to the proximity-based design of the casing, which ensures fixed positioning of the sensor and consistent illumination from the built-in white LED. At distances greater than 0cm, accuracy drops significantly—especially under bright conditions—due to the influence of environmental light and reduced color intensity.

Interestingly, detection at 1cm under dim light performs better (68.75%) than under bright light at the same distance (31.25%). This suggests that the built-in LED is more effective in dim environments, where it is not overpowered by external lighting. Meanwhile, at 2cm, detection reliability is too low to be considered usable.

These findings support the decision to design the device for close-contact usage, with the sensor positioned nearly touching the object. The internal LED improves lighting consistency, but cannot fully compensate for distance-based color shifts. Therefore, for reliable results, users should place the sensor as close as possible to the target surface.

D. Observed Limitations

During the testing process, several limitations were identified in the device’s performance, particularly related to its ability to distinguish certain colour categories and the sensitivity of the sensor to distance and light variation.

One of the most prominent challenges observed was the difficulty in distinguishing grayscale colours, specifically black, white, and grey. These colour often have very similar HSV (Hue, Saturation, and Value) readings when captured by the TCS34725 sensor. For instance, sample readings showed values such as:

TABLE IV. SENSOR HSV READINGS FOR GRAYSCALE

Object Color	H	S	V	Lux	Observation
Black	46	0.38	0.40	46	Low V and Lux
Gray	41	0.40	0.42	636	Slightly higher V and Lux
White	43	0.44	0.42	1195	Higher Lux but similar HSV

Although these colours differ significantly in brightness (value), the similarity in hue and saturation causes the k-NN classifier (trained only on HSV values) to frequently misclassify them, especially when ambient lighting conditions or surface reflectance vary.

Additionally, because the current implementation of the classification algorithm does not consider lux or brightness readings independently, the device lacks a separate mechanism to interpret lightness beyond what is encoded in

HSV. As a result, this can lead to inaccurate identification when encountering neutral tones.

Another limitation is that the sensor's accuracy degrades significantly with increased distance. As shown in the previous section, colour detection at 1cm and 2cm distance drops in performance, with greater shifts in colour interpretation. This highlights the need for precise placement of sensor very close to the object for optimal results. Although a mechanical casing was designed to fix this position at 0cm, any deviation (e.g., due to user movement) may affect results.

Lastly, it is also noted that the device only uses $k = 1$ in the k -NN algorithm, meaning classification is based on the closest single sample in the training data. While this provides quick classification and lower computational burden, it makes the system sensitive to noise or slight reading deviations. Using a higher k could improve stability by averaging multiple neighbours, which is discussed in the suggestion section.

IV. CONCLUSION

This research successfully developed a color recognition system with audio feedback to aid individuals with visual impairments in identifying object colors. Built using an Arduino Nano, TCS34725 color sensor, and DFPlayer Mini, the system applies a simple k -Nearest Neighbor (k -NN) classification algorithm based on HSV color values.

Testing results show that the system performs optimally at a 0cm distance, achieving up to 81.25% accuracy in both bright and dim lighting conditions. However, accuracy significantly declines with increased distance—dropping to 31.25% at 1cm and 12.5% at 2cm under bright lighting. Interestingly, in dim environments, the 1cm accuracy improves slightly to 68.75%, likely due to reduced ambient light interference. These results confirm the device performs most reliably when placed in direct contact with the object surface.

The system faces limitations in distinguishing grayscale colors such as black, white, and gray, due to the similarity of their HSV values. While lux readings from the sensor show potential in helping to separate these colors, they were not considered in the current classification process.

Physically, the device measures approximately 10 x 6 x 5.5 cm and is suitable for handheld use, particularly for close-range object interaction. The built-in white LED enhances consistency across various lighting conditions.

In conclusion, the device offers a promising low-cost and accessible solution for color recognition among visually impaired users. Future improvements may include the integration of lux data into the classification model, expansion of the training dataset, and enhancements for better usability in real-world scenarios.

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DATA AVAILABILITY STATEMENT

Data Availability: Dataset is available from the [Github Repository](#).

OPEN CONTRIBUTORSHIP

Ivan Alexander was responsible for the design and development of the hardware system, including the integration of the TCS34725 sensor with the ESP32 microcontroller. Ivan also supervised the device testing and performance analysis under various environmental conditions.

Marcel Saputra contributed to the design of the business aspects and sustainability analysis of the project, including the evaluation of the materials used for the casing and the device's power saving strategy. Marcel also played a role in the technical documentation and preparation of the research report.

Arizal Ismoyo Wijanarko led the development of the software for sensor data processing and the implementation of the haptic feedback logic. Arizal was also responsible for sensor calibration and system optimization to provide accurate and reliable results.

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